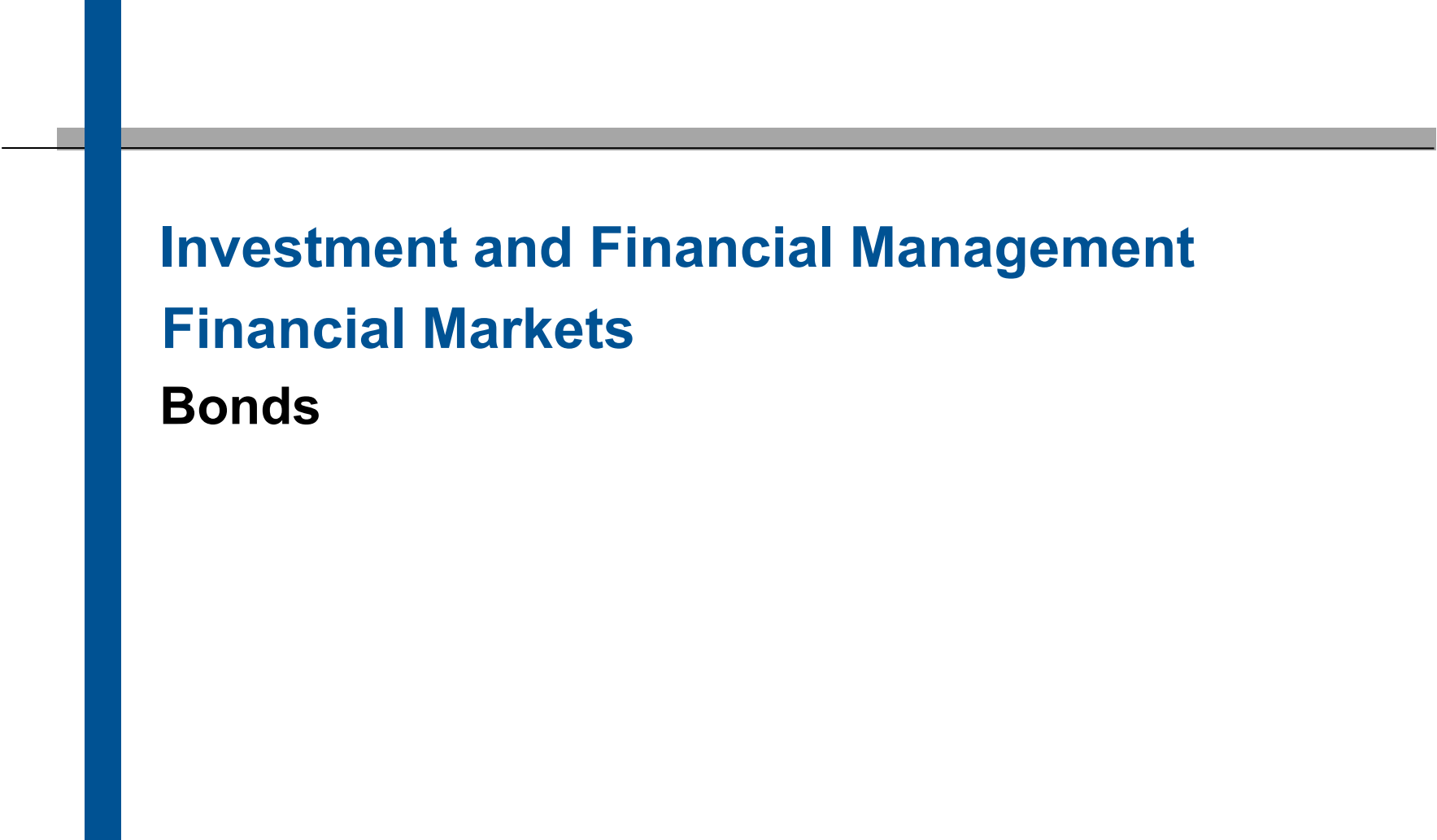




Investment and Financial Management

Financial Markets

Bonds

4.1 Introduction

4.2 Bond pricing

4.2.1 Pricing of zerobonds

4.2.2 Pricing of coupon bonds

4.3 Features of bonds

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4.4 Exercises

Risks associated with changes of the market interest rate

Interest rate risk is the risk that the relevant market interest rate changes.

As a consequence, interest rate changes lead to changes of ...

- the current price of a bond by affecting the relevant discount rate, and
- the conditions for future reinvestments of coupon and redemption payments.

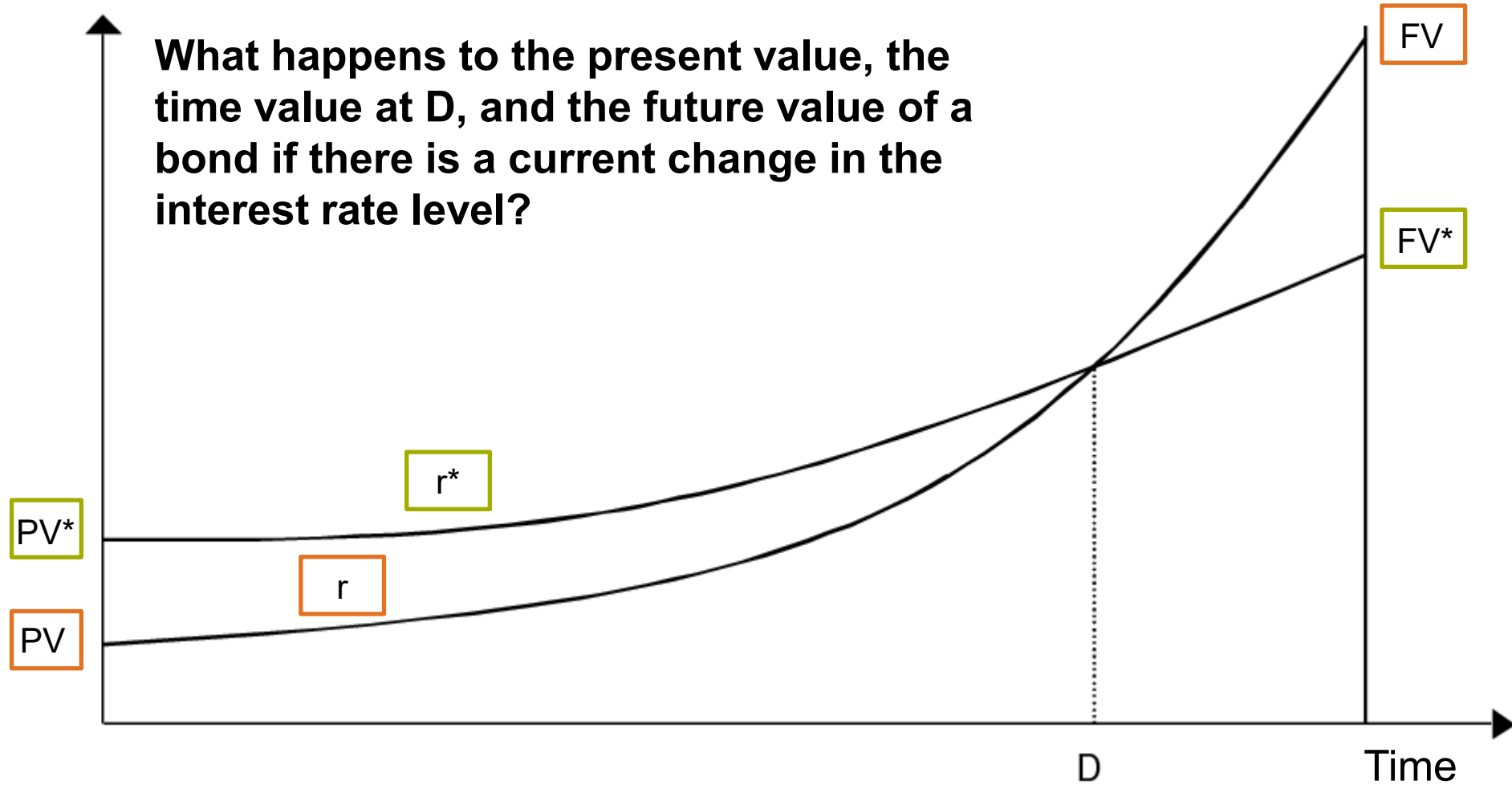
Important: Interest rate changes have a **two-sided, opposite effect on the price and potential reinvestments**. If the interest rate increases, the price of the bond goes down, but potential reinvestments become more attractive.

Risks associated with changes of the market interest rate

Question: Can an investor protect himself against these risks associated with a potential change in the market interest level leading to a change in the value of the bond and affecting the conditions to reinvest?

- Yes. There is a point in time when both effects of interest rate changes offset each other.
- If the investor holds the bond until this point, he is not exposed to any interest rate risk, since the time value of all cash flows at this point in time will not depend on the market interest rate.

What happens to the present value, the time value at D , and the future value of a bond if there is a current change in the interest rate level?



An investor possesses a coupon bond with a face value of 100 EUR, a maturity of five years, and a coupon of 10%. The current market interest rate equals 10%.

Consider two scenarios.

- Scenario 1: New market interest rate instantly after the purchase equals 12%.
- Scenario 2: New market interest rate instantly after the purchase equals 8%.

The investor has three alternatives how to react to the change in the interest rate.

- Alternative 1: Selling the bond after one year.
- Alternative 2: Selling the bond after three years.
- Alternative 3: Holding the bond until maturity.

What are the effective yields for the three alternatives?

An investor possesses a coupon bond with a face value of 100 EUR, a maturity of five years, and a coupon of 10%. The current market interest rate equals 10%.

What are the effective yields for the three alternatives?

Alternative 1: Selling the bond after one year.

$$(1): B_1 = \left[C * \frac{(1 + r_M)^N - 1}{r_M} + B_N \right] * \frac{1}{(1 + r_M)^N} \Rightarrow B_1^{12\%} = 93.93; B_1^{8\%} = 106.62$$

$$(2): r_{eff} = \frac{C + B_1}{100} - 1 \Rightarrow r_{eff}^{12\%} = \frac{10 + 93.93}{100} - 1 = 3.93\%; r_{eff}^{8\%} = \frac{10 + 106.62}{100} - 1 = 16.62\%$$

Alternative 2: Selling the bond after three years.

$$(1): B_3 = \left[C * \frac{(1 + r_M)^N - 1}{r_M} + B_N \right] * \frac{1}{(1 + r_M)^N} \Rightarrow B_3^{12\%} = 96.62; B_3^{8\%} = 103.57$$

$$(2): r_{eff} = \sqrt[3]{\frac{10 * (1 + r_M)^2 + 10 * (1 + r_M) + 10 + B_3}{100}} - 1 \Rightarrow r_{eff}^{12\%} = 9.24\%; r_{eff}^{8\%} = 10.80\%$$

An investor possesses a coupon bond with a face value of 100 EUR, a maturity of five years, and a coupon of 10%. The current market interest rate equals 10%.

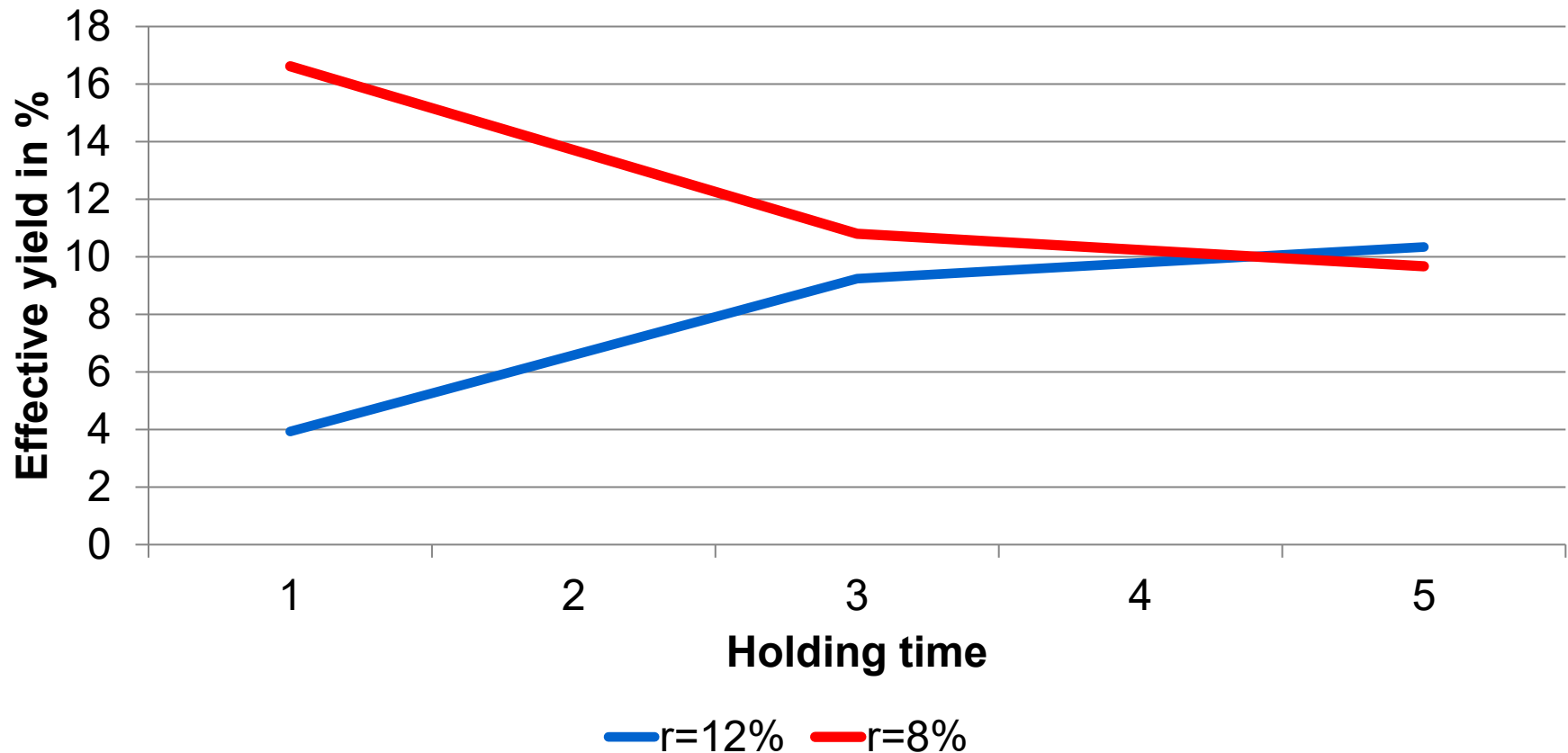
What are the effective yields for the three alternatives?

Alternative 3: Holding the bond until maturity.

$$(1): B_5^{12\%} = 100; B_5^{8\%} = 100$$

$$(2): 100 * (1 + r_{eff})^5 = \sum_{k=0}^4 10 * (1 + r_M)^k + B_5$$

$$(3): r_{eff}^{12\%} = 10.34\%; r_{eff}^{8\%} = 9.67\%$$



Duration: 4.17 years: At this holding period, the effective return is identical for both scenarios and not dependent on any changes in the market interest rate level.

Measure the interest sensitivity of a bond

How strong does the present value of a bond react to any changes in the interest rate?

- *We will see:* The sensitivity can be approximated by the measure of duration.
- The higher the duration, the higher the interest sensitivity.

The concept of duration makes some simplifying assumptions

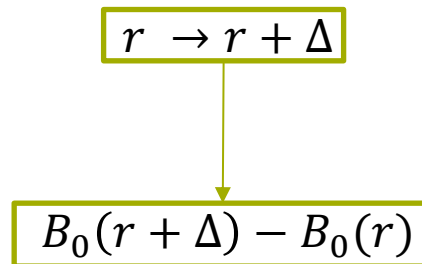
- Flat yield curve
- one-time change of the market interest rate instantly after the purchase of the bond
- Only the level of the yield curve changes, not the slope or the curvature of the curve
- Coupon payments are reinvested with the market interest rate r

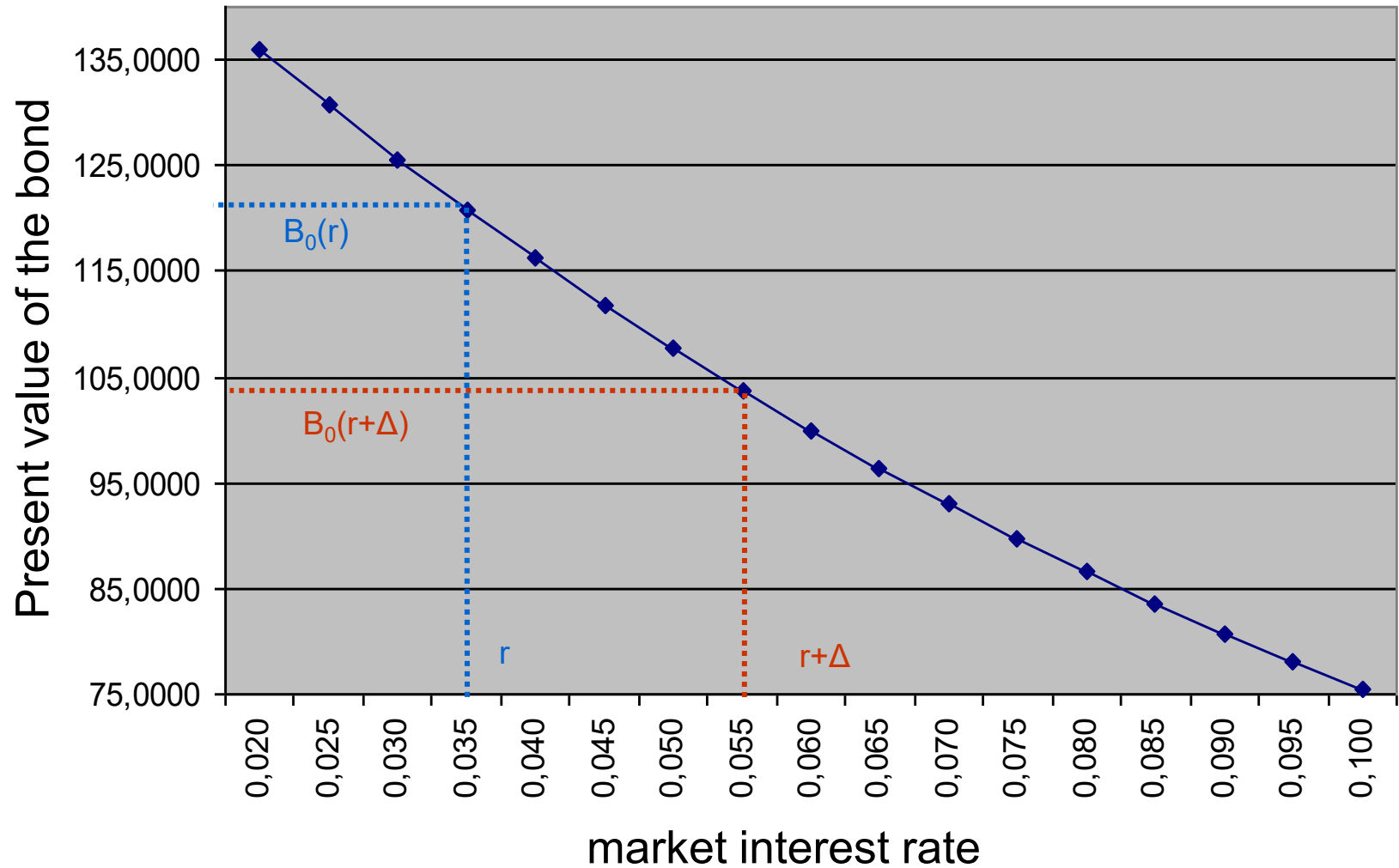
Present value curve as a function of the market interest rate

- Assumption: flat yield curve: $r_1, r_2, r_3, \dots = r$
- Interest rate may vary
- Present value of a bond is described as a function of the market interest rate:

$$B_0 = \sum_{k=1}^N \frac{C}{(1+r)^k} + \frac{B_N}{(1+r)^N} = B_0(r)$$

Let's focus on the impact of a change in the interest level:

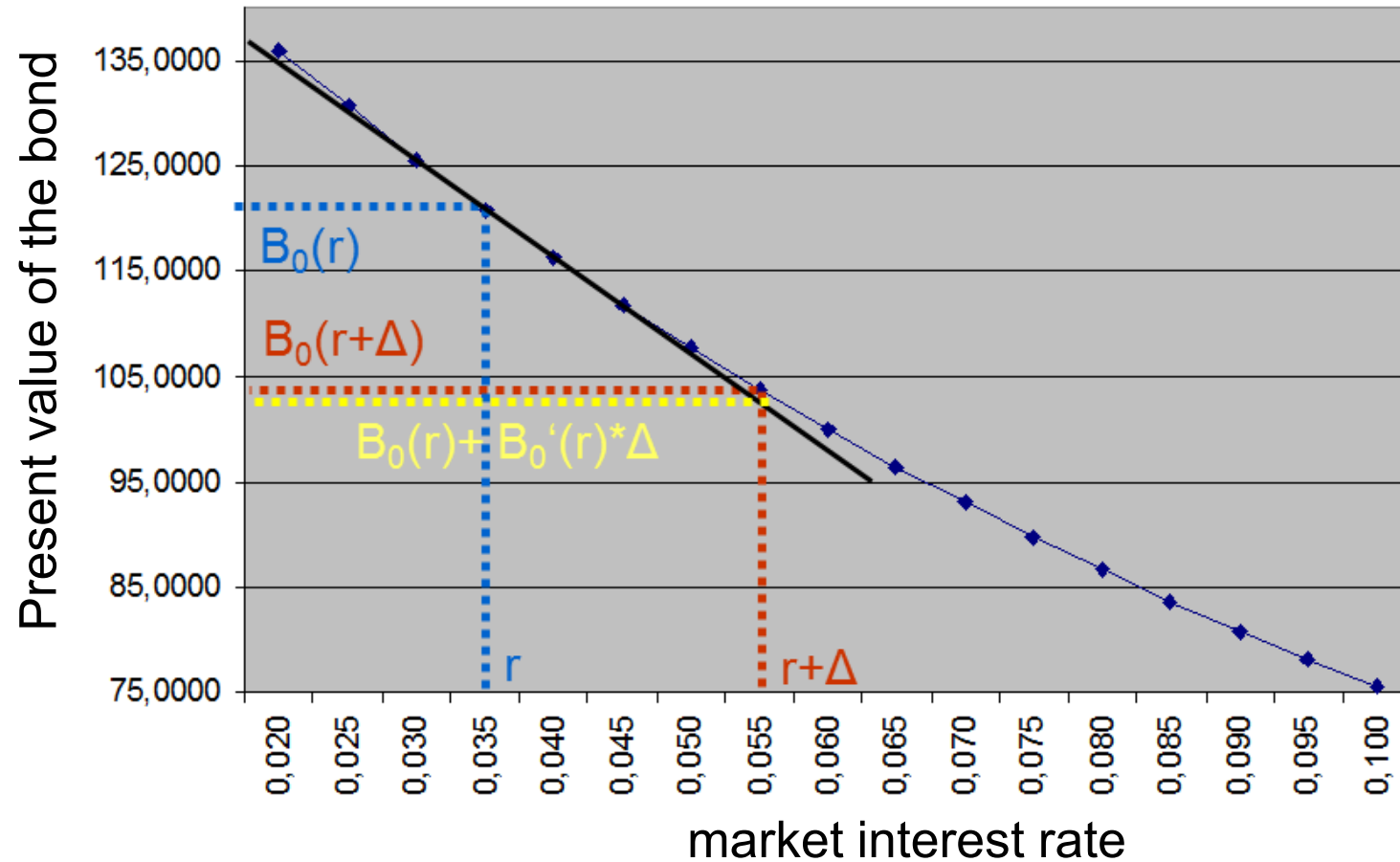




How does the present value of a bond change when the market interest level changes?

- Exact calculation is possible using the proposed formulas
- Difference of the present value between r and $r + \Delta$ represents impact of a change in the interest level and can be approximated in every point of the curve by the product of the slope of the tangent $B_0'(r)$ in the corresponding point r and the difference in the interest rate level Δ
- We will see: Duration gives this tangent and is a good tool for approximation

How does the present value of a bond change when the market interest level changes?



Taylor series: $B_0(r + \Delta) - B_0(r) = B_0'(r) * \Delta + \frac{1}{2} * B_0''(r) * \Delta^2 + \frac{1}{6} * B_0^{(3)}(r) * \Delta^3 + \dots$

Approximate the difference by only using the first term: $B_0(r + \Delta) - B_0(r) \cong B_0'(r) * \Delta$

The present value curve can be approximated by the tangent of the curve in a specific point. The slope of the tangent describes the sensitivity of the curve.

The absolute change in value $B_0(r + \Delta) - B_0(r)$ is a value in money. Often, it is asked for the relative change $\frac{B_0(r+\Delta)-B_0(r)}{B_0(r)}$. This can be approximated by $\frac{B_0'(r)}{B_0(r)} * \Delta$.

Calculation of the derivative:

$$\frac{B_0'(r)}{B_0(r)} = -\frac{1}{1+r} * \left\{ c_1 * \frac{(1+r)^{-1}}{B_0(r)} * \underline{1} + c_2 * \frac{(1+r)^{-2}}{B_0(r)} * \underline{2} + \dots + (c_N + B_N) * \frac{(1+r)^{-N}}{B_0(r)} * \underline{N} \right\}$$

=Duration (D) (also “Macaulay-Duration”)

=Modified Duration (D_{mod})

The duration is the average of the payment dates of a security weighted with the corresponding present value proportion.

$$(1): D = \frac{1}{B_0} * \sum_{k=1}^N \frac{k * C_k}{(1+r)^k} + \frac{N * B_N}{(1+r)^N}$$

The duration represents the **point in time** where there is a **full immunization of the portfolio against potential changes in the market interest rate**. That is why the duration is called the average commitment period.

Calculation of the interest rate sensitivity and the modified duration

Relative interest rate sensitivity

$$(1): \frac{B_0(r + \Delta) - B_0(r)}{B_0(r)} \cong \frac{B'_0(r)}{B_0(r)} * \Delta = -\frac{1}{1 + r} * D * \Delta$$

- What is the price change in %, if there is a 1% change in the interest rate?
- The sensitivity increases with the duration of a bond.
- Increases in the interest rate $\Delta > 0$ lead to value losses, decreases $\Delta < 0$ lead to value gains.

Absolute value change

$$(2): B_0(r + \Delta) - B_0(r) \cong -\frac{1}{1 + r} * D * B_0(r) * \Delta$$

Calculation of the interest rate sensitivity and the modified duration

Modified Duration: Simplification

$$(1): D_{mod} = \frac{1}{1+r} * D$$

$$(2): \frac{B_0 * (r + \Delta) - B_0(r)}{B_0(r)} \cong - D_{mod} * \Delta$$

Concept of duration – Example

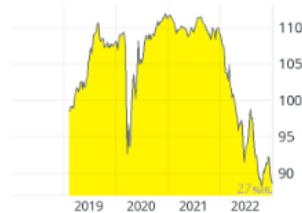
BMW Finance N.V. EO-Medium-Term Notes

2019(29) Euro-Anleihe

WKN: A2RXDR ISIN: XS1948611840



5 year chart ▾



6 month chart ▾



Intraday chart ▾



Course Dates

stock exchange	Stuttgart
Currently	89.74%
time	1/5/23 12:05 PM
differential day before	-0.13%
daily vol.	0.00
ØVol 30T	20,454.55
liquidity	medium
money	89.74
letter	90.34
time	1/5/23 12:27 PM

base data

remaining term	6.1 years
Due date	02/06/2029
issue date	06.02.2019
nominal interest rate	1.500%
bond volume	2 billion
Coupon Art	Firmly
interest payment	Year
interest date	06.02.
issuer	BMW Finance NV
seat emitter	Netherlands
Type	bond
currency	EUR
minimum investment	1,000.00
denomination ⓘ	1,000.00
ISIN	XS1948611840

return metrics

Calculation:	01/05/23 02:00
Yield pa	3.369%
accrued interest	1,385
duration	5.75
Mod. Duration	5,565
convexity	34,318
interest rate elasticity	0.187

News

No entry.

brochures

Key information document not available

Duration of specific instruments

- Duration of a zerobond equals the maturity of the bond since there are no coupon payments
- Duration of coupon bonds
 - The larger, the larger the time to maturity
 - The larger, the lower the market interest rate
 - The larger, the lower the coupon
- Duration of a perpetual (this implies that with increasing time the duration converges to a limit value; the impact of the time on the duration decreases)

$$(1): D^{perp} = \frac{1 + r}{r}$$

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 - 4.3.3 Concept of duration**
- 4.4 Exercises**

(A.4) Two bonds with a face value of 100 EUR are traded.

- A: $C = 6\%$ with a repayment at premium of 101% and $N = 3$.
- B: $C = 12\%$ with a repayment at face value and $N = 6$.

Please calculate the duration of the bonds A and B and assume a market interest rate of $r = 9\%$.

Duration of Bond A:

$$(1): B_0 = \sum_{k=1}^N \frac{C}{(1+r)^k} + \frac{B_N}{(1+r)^N} = \frac{6}{1.09} + \frac{6}{1.09^2} + \frac{107}{1.09^3} = 93.1783$$

$$(2): D^A = \frac{1}{B_0} * \sum_{k=1}^N \frac{k * c_k}{(1+r)^k} + \frac{N * B_N}{(1+r)^N} = \frac{1}{93.1783} * \left(\frac{1 * 6}{1.09} + \frac{2 * 6}{1.09^2} + \frac{3 * 107}{1.09^3} \right)$$

$$(3): D^A = \frac{263.4756}{93.1783} = 2.83$$

**How to interpret
the result?**

(A.4) Two bonds with a face value of 100 EUR are traded.

- A: $C = 6\%$ with a repayment at premium of 101% and $N = 3$.
- B: $C = 12\%$ with a repayment at face value and $N = 6$.

Please calculate the duration of the bonds A and B and assume a market interest rate of $r = 9\%$.

Duration of Bond B:

$$(1): B_0 = \sum_{k=1}^N \frac{C}{(1+r)^k} + \frac{B_N}{(1+r)^N} = \frac{12}{1.09} + \frac{12}{1.09^2} + \frac{12}{1.09^3} + \frac{12}{1.09^4} + \frac{12}{1.09^5} + \frac{112}{1.09^6} = 113.4578$$

$$(2): D^B = \frac{1}{B_0} * \sum_{k=1}^N \frac{k * c_k}{(1+r)^k} + \frac{N * B_N}{(1+r)^N}$$

$$(3): D^B = \frac{1}{113.4578} * \left(\frac{1 * 12}{1.09} + \frac{2 * 12}{1.09^2} + \frac{3 * 12}{1.09^3} + \frac{4 * 12}{1.09^4} + \frac{5 * 12}{1.09^5} + \frac{6 * 112}{1.09^6} \right)$$

$$(4): D^B = \frac{532.7000}{113.4578} = 4.70$$

(A.5) A zerobond is traded with a maturity of ten years. The current market interest rate equals 6%. We will look at possible changes of the interest rate of 2%. Use the duration to identify potential value gains and value losses and compare your results with the real gains or losses.

$$(1): D^{ZB} = 10$$

$$(2): D_{mod}^{ZB} = \frac{1}{1+r} * D^{ZB} = \frac{10}{1.06} = 9.43$$

Relative approximated value change using the duration

$$(3): \frac{B_0 * (r + \Delta) - B_0(r)}{B_0(r)} \cong - D_{mod} * \Delta \cong - 9.43 * \pm 2\% \cong \pm 18.86\%$$

(A.5) A zerobond is traded with a maturity of ten years. The current market interest rate equals 6%. We will look at possible changes of the interest rate of 2%. Use the duration to identify potential value gains and value losses and compare your results with the real gains or losses.

Real value change (no approximation)

$$(1): r = 4\% \rightarrow B_0^{ZB} = \frac{100}{1.04^{10}} = 67.56$$

+20.99% vs. +18.86%

$$(2): r = 6\% \rightarrow B_0^{ZB} = \frac{100}{1.06^{10}} = 55.84$$

-17.05% vs. -18.86%

$$(3): r = 8\% \rightarrow B_0^{ZB} = \frac{100}{1.08^{10}} = 46.32$$

The example shows that the approximation using the duration is quite accurate. However, it also shows that value losses are overestimated, gains are underestimated.

(A.5) A zerobond is traded with a maturity of ten years. The current market interest rate equals 6%. **We will look at possible changes of the interest rate of 4%.** Use the duration to identify potential value gains and value losses and compare your results with the real gains or losses.

$$(1): D^{ZB} = 10$$

$$(2): D_{mod}^{ZB} = \frac{1}{1+r} * D^{ZB} = \frac{10}{1.06} = 9.43$$

Relative approximated value change

$$(3): \frac{B_0 * (r + \Delta) - B_0(r)}{B_0(r)} \cong - D_{mod} * \Delta \cong - 9.43 * \pm 4\% \cong \pm 37.72\%$$

(A.6) A zerobond is traded with a maturity of ten years. The current market interest rate equals 6%. **We will look at possible changes of the interest rate of 4%.** Use the duration to identify potential value gains and value losses and compare your results with the real gains or losses.

Real value change (no approximation)

$$(1): r = 2\% \rightarrow B_0^{ZB} = \frac{100}{1.02^{10}} = 82.03$$

+46.91% vs. +37.72%

$$(2): r = 6\% \rightarrow B_0^{ZB} = \frac{100}{1.06^{10}} = 55.84$$

-30.96% vs. -37.72%

$$(3): r = 10\% \rightarrow B_0^{ZB} = \frac{100}{1.10^{10}} = 38.55$$

The example shows that the approximation becomes less accurate when the interest rate change increases.

Evaluation of the duration approach

- A number of assumptions necessary
 - But: Assumption of flat yield curves could be relaxed
- Estimation is always rather pessimistic than optimistic
 - Overestimation of value losses, underestimation of value gains
 - Effect is even stronger when the potential interest change increases

If the linear approximation does not seem sufficient, we would need to include the second term of the Taylor series (Concept of Convexity).

Duration of a portfolio

The duration of a portfolio can be computed easily, following two steps:

1. Determine the duration of the single securities D_i
2. Calculating the portfolio duration D_{PF} as value-weighted (x_i) sum of the durations of the single securities

$$(1): D_{PF} = \sum_{i=1}^N x_i * D_i; \text{ with } x_i = \frac{w_i}{\sum_{i=1}^N w_i}$$

(A.7) A portfolio should be constructed using bond A ($D_A = 2.8277$) and bond B ($D_B = 4.6951$) from A.4 which should be immunized against potential interest rate changes at $t=4$ ($D = 4$). Please calculate the fractions of bond A and bond B which need to be used.

$$(1): D_{PF} = \sum_{i=1}^N x_i * D_i$$

$$(2): x_A * 2.8277 + x_B * 4.6951 = 4$$

$$(3): x_A + x_B = 1$$

$$(4): (1 - x_B) * 2.8277 + x_B * 4.6951 = 4$$

$$(5): x_B = 0.6278; x_A = 0.3722$$

(A.8) An investor wants to invest 10,000 EUR so that she is immune against potential changes to the market interest rate at $t=4$. The current market interest rate equals 10%, the yield curve is flat.

(a) Which amount does she need to invest in a one-year and a seven-year zerobond if she wants her portfolio to be immunized at $t=4$.

$$(1): D_{PF} = \sum_{i=1}^N x_i * D_i = \text{investor's planning horizon}$$

$$(2): x_A * 1 + x_B * 7 = 4$$

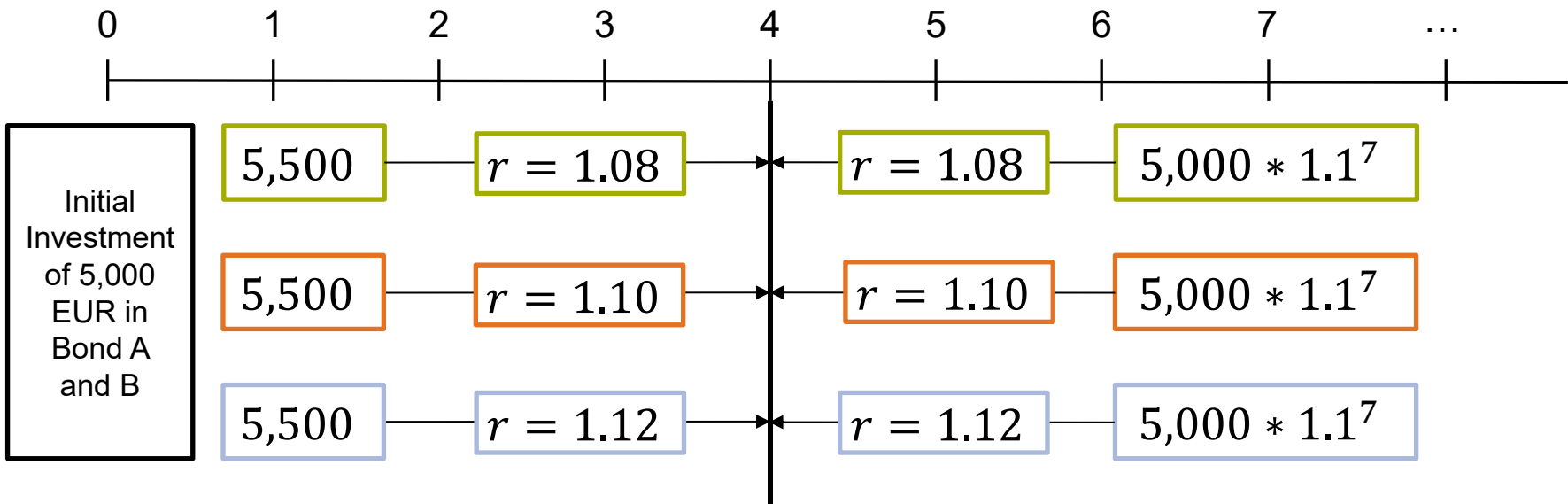
$$(3): x_A + x_B = 1$$

$$(4): x_B = 0.5; x_A = 0.5$$

The investor needs to invest 5,000 EUR in bond A and bond B.

(A.8) An investor wants to invest 10,000 EUR so that he is immune against potential changes to the market interest rate at $t=4$. The current market interest rate equals 10%, the yield curve is flat.

(b) What is the investor's wealth in $t=4$ if the interest rate remains unchanged? How does his wealth differ if the interest rate drops to 8% or goes up to 12%?



(A.8) An investor wants to invest 10,000 EUR so that he is immune against potential changes to the market interest rate at $t=4$. The current market interest rate equals 10%, the yield curve is flat.

(b) What is the investor's wealth in $t=4$ if the interest rate remains unchanged? How does his wealth differ if the interest rate drops to 8% or goes up to 12%?

$$(1): r = 10\% \rightarrow V_4 = 5,500 * 1.1^3 + 5,000 * \frac{1.1^7}{1.1^3} = 14,641.00 \text{ EUR}$$

$$(2): r = 8\% \rightarrow V_4 = 5,500 * 1.08^3 + 5,000 * \frac{1.1^7}{1.08^3} = 14,663.18 \text{ EUR}$$

$$(3): r = 12\% \rightarrow V_4 = 5,500 * 1.12^3 + 5,000 * \frac{1.1^7}{1.12^3} = 14,662.40 \text{ EUR}$$

Portfolio is immune against changes in the interest rate level. Small differences come from the linear approximation.